

AKANKSHA BOLLA

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SUMMARY

Computer Science Master's candidate and AI builder with hands-on experience developing AI enabled software using Python, React, and FastApi. Skilled at applying modern AI tooling large language models, agentic workflows, prompt design, and retrieval alongside core computer science fundamentals such as data structures, system design, and design patterns. Routinely uses AI coding assistants to accelerate development while keeping code readable, testable, and maintainable. Seeking a Software Engineer role on a AI-focused team.

TECHNICAL SKILLS

Languages: Python, JavaScript, Core Java, HTML, CSS

AI / ML & GenAI: LLMs, Langchain, Langraph , Agentic Workflows, Fine tuning, Prompt Engineering, RAG pipelines, Hugging Face Transformers, BERT, Scikit-learn, Word2Vec, ML pipelines, PyTorch, edge deployment

Frameworks & Tools: React.js, Flask, Spring Boot, Angular, Node.js, REST APIs, FastApi, Git, Postman,

Databases & Cloud: MySQL, AWS, Docker, cPanel

PROJECTS

Learnova – AI-Powered Learning Application ([Live Demo](#))

- Designed and deployed an AI-powered study assistant supporting 3 learning modes: summarization, flashcard generation, and practice Q&A.
- Integrated Hugging Face NLP models and Gemini APIs with custom prompt engineering, reducing study preparation time by ~50% in user testing.
- Implemented modular prompt templates that dynamically adapt to different subjects and difficulty levels, improving response relevance.

EmotiveChat – Emotion-Aware AI Chatbot ([Github](#))

- Built a real-time sentiment-aware chatbot that detects 7 emotional states using a fine-tuned BERT model with 89% classification accuracy.
- Integrated BlenderBot for generative, context-aware responses, improving user satisfaction scores by 35%.
- Designed a clean React.js frontend with real-time emotion tagging displayed alongside chat responses.

Food Calorie Estimation System ([Github](#))

- Developed an AI-powered food calorie estimation system using Computer Vision and Machine Learning to identify food items from images and estimate nutritional values.
- Implemented image preprocessing, feature extraction, and food classification pipelines using Python, OpenCV, and deep learning techniques.

PROFESSIONAL EXPERIENCE

Texas A&M University *Research Assistant*
Corpus Christi, Texas

Jan 2025 – May 2026

- Engineered a Generative AI pipeline to automate test case generation from requirements and user stories, reducing manual effort by an estimated 40%.
- Designed and iterated on prompt templates to produce structured, consistent test-case outputs aligned with SDLC validation requirements.
- Implemented automated validation checks on AI-generated outputs, improving clarity and requirement alignment by 30%.

Secure Info Tech *Data Analyst*
Remote, Bengaluru, India

May 2023 – June 2024

- Developed automated data preprocessing and transformation pipelines using Python, Pandas, and SQL for large-scale datasets.
- Performed exploratory data analysis and feature engineering to support machine learning and predictive analytics initiatives.
- Optimized SQL queries and data workflows, improving data retrieval efficiency and reducing processing time.
- Built dashboards and analytical reports that translated complex datasets into actionable business insights for stakeholders.

EDUCATION

Texas A&M University **Aug 2024- May 2026**

Master's, Computer Science | GPA 3.4/4.0

GITAM University **Sep 2020- May 2024**

B.Tech, Computer Science | GPA 3.8/4.0

CERTIFICATIONS

- AWS Academy Cloud Foundations – Eduskills
- Introduction to Artificial Intelligence – IBM
- Python and Django Full Stack Web Development – Udemy